



Corporate Profile



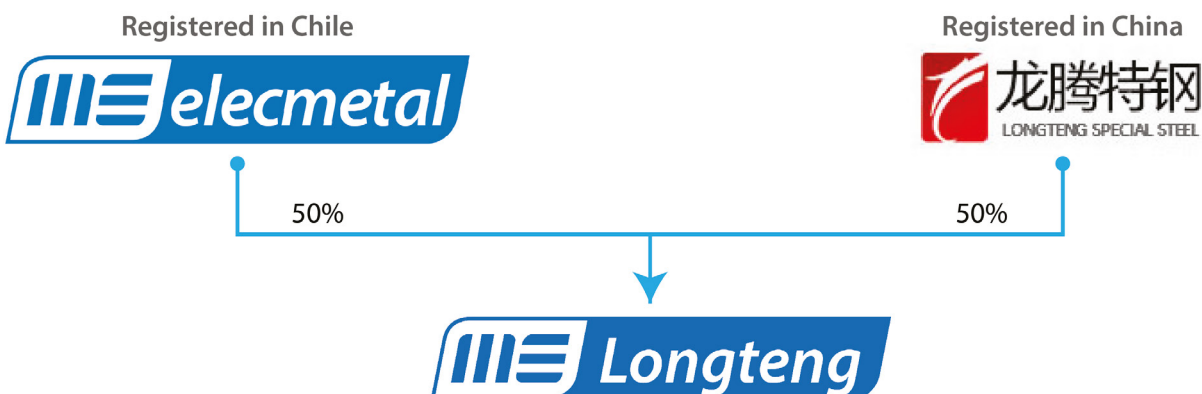
Introduction to ME Long Teng Grinding Media

ME Long Teng Grinding Media is a joint venture between ME Elecmetal and Longteng Special Steel Co., Ltd — two worldwide leaders in the production and manufacture of steel consumable parts for the mining industry.

ME Elecmetal is a public company traded on the Santiago Exchange in Chile. The majority of company shares are owned by Claro Group, one of the most prominent conglomerates in South America. ME Elecmetal is currently the largest global mill liner supplier.

Longteng Special Steel Co., Ltd. is an integrated steel mill based in China, leading globally in the niche markets of several specialized steel products.

- Our facility in China is the world's largest single site grinding ball manufacturing plant by capacity: 500,000+ MT
- Our plant in Zambia is the most technologically advanced ball manufacturing facility in Africa, with over 100,000 MT installed capacity
- Industry leading equipment design and process control, certified by ISO 9001, ISO 14001 and OHSAS18000
- State-of-the-art lab and testing capabilities with the world's tallest drop ball machine



Corporate Profile

The Elecmetal Group

- ME Elecmetal is a part of The Elecmetal Group with over USD one billion per annum in revenues
- ME Elecmetal is a leading wear consumable supplier for mining, construction, and energy, with net revenues over USD 600 million per annum
- Range of products include: grinding media, mill liners, crusher liners and GET
- Range of services include: service centers, engineering and design, Early Alert, ME FIT System™ and ME UpTime+
- 1,000+ employees
- Customers in 35+ countries around the world
- Manufacturing capacity of 134,500+ tons of mill liners in five foundries facilities
- Manufacturing capacity of 600,000+ tons in grinding media in our joint venture facilities

The Elecmetal Group



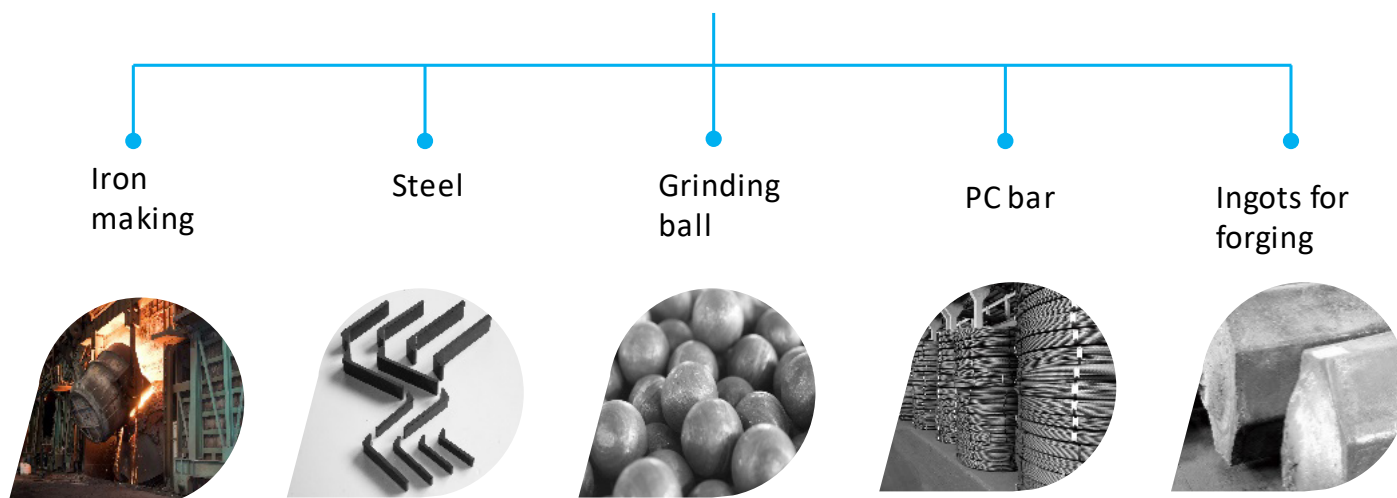
Corporate Profile



The Longteng Group

- The Longteng Group was established in 1993
- Covers an area of over 1.6 million square meters with a staff of 5000+
- Four main products series (Three million metric tons per year)
 - Steel ball series for bearing & mining industry
 - Ship building section
 - High-strength, pre-stressed concrete steel bars
 - Forging ingot series

The Longteng Group



Vision, Mission & Values

Our Vision

Our vision is to be a globally competitive supplier, recognized by its excellence and leadership in the delivery of integral solutions which add value to the mining processes and to other target markets.

Our Mission

Our mission is to meet the needs and exceed the expectations of our shareholders, customers, employees and communities where we operate, by providing comprehensive solutions to the market.

Our Values

- Respect of dignity
- Teamwork
- Commitment to the community and environment
- Creativity and innovation
- Customer focus
- Proactive search for excellence
- Human development
- Responsibility and integrity

Operating History & Production Capacity

Operating History

As a part of the development of new products and markets, and our vision to deliver integral solutions to our clients, ME Long Teng has developed technology, metallurgical as well as process, for the manufacture of superior quality forged steel balls.

Through exclusive manufacturing agreements with Changshu Longteng Special Steel Co., Ltd., ME Elecmetal's new products "ME Super SAG" and "ME Ultra Grind" began to be commercialized worldwide in 2009. In 2011, ME Elecmetal and Changshu Longteng Special Steel Co., Ltd. incorporated, in China, to create the 50/50 joint-venture company, "ME Long Teng Grinding Media (Changshu) Co. Ltd." ("ME Long Teng"), for the manufacturing of these products sold by ME Elecmetal worldwide. Subsequent to the launching of a pilot plant in 2010, ME Long Teng initiated in 2011 the construction of a modern manufacturing plant for grinding balls, in the city of Changshu, China, with a total capacity of production of about 500,000 Metric Tons of grinding balls, which ME Elecmetal directly commercializes worldwide. In 2018, the fifth phase of this project was finalized.

Grinding Media Production Capacity

Currently, the Joint Venture facility in China has an annual production capacity over 500,000 MT of forged steel balls, and 60,000 MT of grinding rods. The annual production capacity of the facility in Zambia is over 100,000 MT of forged steel balls.

In addition to our facility in China and Zambia, ME Long Teng continues to evaluate options for establishing more manufacturing facilities in other parts of the world, in line with our business growth.

Production, Supply Chain & Distribution Channels

Production Site Locations

Our joint venture forged ball plant locations are as follows:

- Jiangshu, China, within 2 hours drive to Shanghai China — is one of the largest ports in the world.
- Kalumbila, Zambia in the Northwest Province of Zambia and within a few hours drive of many of the largest mines in Zambia.

Due to manufacturing flexibility at our ball plants and a wide spectrum of steel making capability at our upstream steel mills, we are proud to be able to customize our products' design and production for our major customers.

Standard sizes produced from normal production

China

- Forged Balls: 20mm – 160mm

Zambia

- Forged balls: 60mm – 80mm and 100mm – 160mm

Additional sizes to be added as the production facility is expanded

Supply Chain Management and Global Distribution Channels

Our grinding media business operates on all continents where there are major mining activities. The company has a team of professionals specialized in global logistics and inventory management to serve customers around the world. Our logistics team has proven experience in establishing solid supply arrangements to ensure continuous grinding media supply to its customers without interruption.

Technical Support

1. Our frontline sales engineers and service staff for both grinding balls and mill liners will provide local support as needed.
2. Our central technical team will back up our local personnel in providing further technical support to our customers as needed.
3. For on-going ball quality monitoring, we will assist our customers through the following:
 - a. Mill charge trajectory modeling, if needed. This way, we ensure a good compatibility between our liner design and ball design to assist our customers in achieving maximum mill throughput and minimal mill liner and ball impact.
 - b. On-going discussion and review of mill operation parameters (ball charge levels, mill speeds, etc) to identify any need for further product design changes for mill liners or grinding balls.
 - c. Scheduled review and discussion with our customers regarding daily ball charge, ball consumption (kg/ton ore grind, kg/Kwh power, etc), ball charge condition in the mill, and ball discharge condition out of pebble ports.
 - d. We share our factory ongoing "pass-fail" drop test results with our customers for the products we supply.
 - e. Continuous improvement on grinding media and mill liner quality as well as application efficiency.

General Warranty

We warrant our balls to have less than 1% breakage under normal wet milling operation condition without operational upsets. Breakage is defined as half balls with evidence of hydrogen, centerline segregation, or porosity.



Quality Assurance Procedures



It is critical for ME Long Teng to maintain a high level of quality and consistency in its products. The following steps are fully executed and documented throughout the manufacturing process.

A. Metallurgical Design

Based on specific application requirements at customer's site, ME Long Teng designs the metallurgical properties of its grinding media accordingly to ensure high level of compatibility between the grinding ball product and the application requirement at customers' site. Overall low wear rate is the end goal. Such material designs must always be kept confidential.

B. Raw Material

By using "clean steel" technology, ME Long Teng grinding balls have ultra-low impurity levels in the alloy raw material, which ensures best final product quality. ME Long Teng specifications have the lowest detrimental chemistry levels in the industry. To ensure the continuity of such competitive advantage, the following quality control procedures must be followed with regards to raw material management:

- The steel material designed by ME Long Teng is only sourced from the Long Teng steel mill that has state of the art steel making equipment and processes. Quality audits are carried out regularly. Use or purchase of steel material outside of the prequalified steel mills must be pre-approved by the President of ME Long Teng in charge of technical and engineering function.
- All steel material must come with a quality certificate for each heat lot indicating conformity to the specifications.
- For ME Super SAG balls material, S and P levels shall be maintained below 0.022%.
- For ME Ultra Grind balls or ME Performa material, S and P levels shall be maintained below 0.025%.
- All steel materials, upon receipt at the ball plant, shall be randomly sampled at 1-5% (depending on the actual need) for compliance in chemistry, segregation, and metallurgical requirement to ensure quality consistency at the ball plant lab.

Quality Assurance Procedures



C. Production Batch management

Based on actual production need, customer order, and plant loading, ball production for a ME Super SAG batch can be established in the range of 100 MT to 1000 MT each batch. Each batch is assigned to one batch serial number, which shall be linked to the following:

1. Bar heat number (traceability back to raw material)
2. Quantity of each bar heat
3. Heat treatment records for the ball batch production
4. All recordable instrumentation reading during the production

Since ME Ultra Grind ball production is a continuous process (different from ME Super SAG ball's batch process), Ultra Grind production and material records shall be monitored and maintained on a continuous time-based process basis without being assigned a production batch serial number.

D. Heat treatment

The proprietary heat treatment processes and procedures at ME Long Teng must always be kept confidential. Variance at each heat treatment stage must be controlled within plus or minus 20 degrees Celsius of the designed temperature.

All heat treatment process data shall be regularly collected, maintained, and sent to ME Long Teng for statistics production control analysis and no products with heat treatment cracks are allowed to be delivered or shipped.

E. Forging

Due to the ME Long Teng high automated forging processes, the roundness of the grinding media is excellent, with very small tolerance of diameter or weight differences. Both the rolling line and forging line are done at high temperature

Quality Assurance Procedures



F. Physical and Chemistry Tests

Following physical tests shall be conducted on 1% samples randomly collected from each production batch, before products are qualified to be shipped.

1. Hardness section and profiling (1 to 20 balls from each batch, depending on the batch size, lab work schedule, product maturity, and customer requirement). Hardness acceptance criteria are based on product specification.
2. Lab chemistry inspection if required by customers (1 to 20 balls from each batch, depending on the batch size, lab work schedule, product maturity, and customer requirement). Chemistry acceptance criteria are based on product specification.
3. Drop ball testing (1 to 20 balls from each batch, depending on the batch size lab work schedule, product development stage, and customer requirement) Ball sample passing drop test simulating one-hour SAG mill grind out without half ball breakage is considered acceptable.
 - a. Ball sample showing half ball breakage in drop test simulating one-hour SAG mill grind out is considered not acceptable. If breakage is found at ball drop test, ball sampling size is required to be tripled, and then quadrupled. If over half of the balls sampled show breakage in the drop test, the whole batch shall be rejected.
 - b. Ball spalling during drop test simulating one-hour SAG mill grind out without half ball breakage is considered acceptable. However, excessive spalling during test needs to be reported, as it is an indicated of potential need for ball design adjustment.

G. Visual & Surface Inspection

All following visual tests shall be conducted on samples randomly collected at 5% of the quantity of each production batch, before products are shipped.

- ✓ Ball weight (nominal with plus or minus 10% allowed)
- ✓ Surface quality (balls samples with obvious forging marks, forging laps, or forging belts shall be within 10% of the balls sampled)
- ✓ Instruction from specific customer, if any, shall be followed.

H. Package & Shipment

The production serial number for the product shall be written on the top of each bulk bag to ensure full traceability back to plant records.

Delivery will be made according to the customers requirements.



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